

Experiment 7: Breadth First Search (BFS) in Python

Aim

To write a Python program to implement the **Breadth First Search (BFS)** algorithm.

Algorithm

1. Create a graph using an adjacency list.
2. Start from the source node.
3. Mark the source node as visited.
4. Visit all adjacent unvisited nodes using a queue.
5. Repeat until the queue becomes empty.
6. Print the BFS traversal.

Program

```
graph = {  
    'A': ['B', 'C'],  
    'B': ['D', 'E'],  
    'C': ['F'],  
    'D': [],  
    'E': [],  
    'F': []  
}
```

```
visited = []
```

```
queue = ['A']
```

```
while queue:
```

```
    node = queue.pop(0)
```

```
    if node not in visited:
```

```
        print(node, end=" ")
```

```
visited.append(node)
queue.extend(graph[node])
```

Input

Starting Node = A

Output

```
      A
     /\
    B  C
   /\ /\
  D E F G

BFS Traversal:
A B C D E F G
```

Result

Thus, the Python program to implement the **Breadth First Search (BFS)** algorithm was executed successfully, and the BFS traversal of the graph was displayed.